

Predictive Warning Detection for Gradual & Sudden Sensor Changes using GRU + MLP Model

1. Summary

This project develops a predictive model capable of detecting early warning signs in sensor data such as gradual increases in vibration or sudden changes in flow rate or pressure. The system uses a combination of Gated Recurrent Unit (GRU) networks and Multi-Layer Perceptrons (MLP) to process historical time-series data and identify patterns that require quick supervisory review.

2. Objective

- Detect gradual vibration increases.
- Detect sudden pressure or flow rate changes within normal operating range.
- Provide early alerts for supervisor attention before failures occur.

3. Data Description

- **Source:** Simulated dataset (final_simulated_sensor_data.csv)
- **Size:** ~20,000 timesteps, recorded every minute.
- **Features:**
 - pressure_psi
 - flow_rate_bbl_day
 - vibration_mm_s
 - Engineered features: pressure_diff, flow_rate_pct_change, vibration_diff
- **Target:** warning_class (0 = normal, 1 = needs attention)

4. Methodology

-> Data Preprocessing:

- Added features for differences & % changes.
- Filled missing values after feature engineering.
- Standardized data for model training.

-> Sequence Creation:

- Used 10-step rolling windows to predict the next state.

-> Model Architecture:

- GRU (64 units) → captures time patterns like gradual trends.
- Batch Normalization → stabilizes training.
- Dense (32 units, ReLU) + Dropout → learns non-linear interactions, prevents overfitting.
- Output Dense (1 unit, Sigmoid) → binary classification.

-> Training:

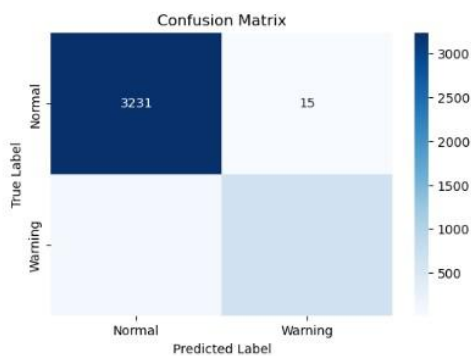
- Train/test split: ~80/20.
- Loss function: Binary Crossentropy.
- Optimizer: Adam.
- Epochs: 20, Batch size: 64.

5. Results

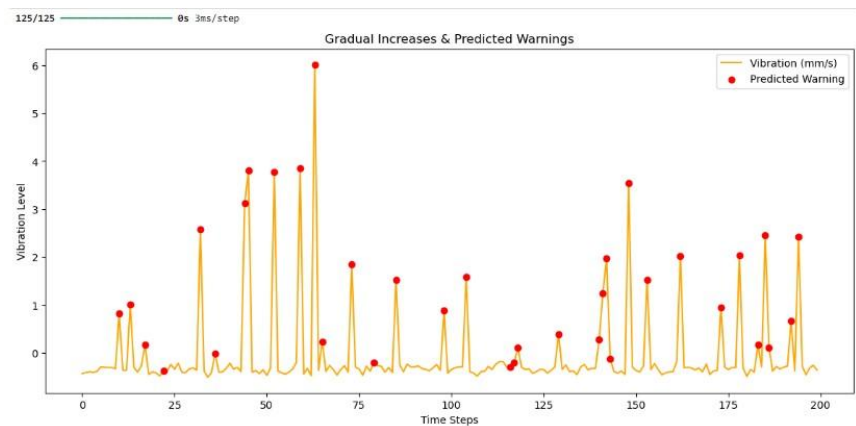
-> Classification report:

Class	Precision	Recall	F1-score	Support
Normal (0)	0.97	1.00	0.98	3246
Warning (1)	0.98	0.88	0.93	752
Overall Accuracy	0.97	-	-	3998

-> Confusion Matrix:



-> Trend Visualization:



-> Alert Message:

```
1/1 ----- 0s 45ms/step
1/1 ----- 0s 39ms/step
1/1 ----- 0s 42ms/step
1/1 ----- 0s 36ms/step
1/1 ----- 0s 41ms/step
1/1 ----- 0s 41ms/step
1/1 ----- 0s 40ms/step
1/1 ----- 0s 41ms/step
1/1 ----- 0s 38ms/step
1/1 ----- 0s 40ms/step
1/1 ----- 0s 42ms/step
⚠ ALERT: Potential issue detected!
Vibration: 0.82 mm/s | Pressure: 0.63 psi | Flow: 0.96 bbl/day
Risk Score: 99.9%

1/1 ----- 0s 40ms/step
1/1 ----- 0s 41ms/step
1/1 ----- 0s 40ms/step
⚠ ALERT: Potential issue detected!
Vibration: 1.01 mm/s | Pressure: 0.28 psi | Flow: 1.64 bbl/day
Risk Score: 99.8%

1/1 ----- 0s 39ms/step
1/1 ----- 0s 41ms/step
1/1 ----- 0s 40ms/step
1/1 ----- 0s 40ms/step
⚠ ALERT: Potential issue detected!
Vibration: 0.17 mm/s | Pressure: -0.38 psi | Flow: 0.55 bbl/day
Risk Score: 100.0%

1/1 ----- 0s 41ms/step
1/1 ----- 0s 40ms/step
1/1 ----- 0s 41ms/step
1/1 ----- 0s 41ms/step
1/1 ----- 0s 40ms/step
⚠ ALERT: Potential issue detected!
Vibration: -0.37 mm/s | Pressure: 7.73 psi | Flow: 1.45 bbl/day
Risk Score: 62.4%
```

6. Interpretation

- **Gradual Changes:** GRU captures slow upward trends in vibration_diff.
- **Sudden Changes:** pressure_diff and flow_rate_pct_change highlight abrupt shifts.
- **Early Warning Capability:** Model can signal issues while still within operating limits.

7. Limitations & Next Steps

- Real-world deployment will require real sensor data retraining.
- Suggest integrating with live dashboard alerts.

8. Conclusion

- The GRU+MLP model meets the stated requirement and can act as an early warning system for supervisors.
- Implementation could reduce downtime and prevent major incidents.